

Referee report: “Sorting, Price Informativeness, and Information Acquisition in Dark Markets” (JEDC-D-26-00041)

This paper presents experimental evidence on information acquisition in dark markets. This is an interesting topic. To my best knowledge, there are two empirical works, and one closely related experimental work. Compared to the experimental work by Asparouhova et al. (2024), the current paper uses a market structure which more closely resembles the practice of dark pool trading. For this, I believe, the paper can help us learn something new. Having said so, the current manuscript lacks theoretical discussions, fails to explain the experimental design in detail and fail to link the findings to theory and existing evidence. Furthermore, some additional analysis may be required. To sum up, it needs significant improvements to meet the journal’s publication standards. Below I provide some suggestions in detail.

General Comments:

1. **Contributions:** The main contribution is to allow information acquisition in experimental dark markets and examine how the dark pool trading would affect information acquisition; and if so, whether improve or deteriorate price informativeness. It relates to two empirical papers: Brogaard and Pan (2023, US market) and Ekanayake et al., (2024, Germany market). Both find dark pool trading facilitates/aids information acquisition. The current paper essentially provides some consistent experimental evidence, in supportive of the empirical findings.
 - 1.1 On the relationship between dark trading and price efficiency, Hendershott et al. (2024) provides the experimental evidence. Hence, I do not suggest the authors to position the paper from this angle. But the authors need to cite the work and discuss any related findings - also lower the priority of presenting similar findings to theirs.
 - 1.2 The paper should place strong focus on the relationship between information acquisition and price informativeness. The current manuscript provides ALL results and fails to highlight to readers the key ones. In my opinion, subsection D (on page 30) essentially is the research question and the corresponding findings. But it is currently buried among others. Additional analysis would be required to make this section stronger and richer so that we can learn from the experiments. Results in other sections that are consistent with the existing literature can be reported briefly or in Online Appendix.
 - 1.3 The paper is also closely related to Asparouhova et al. (2024) which has an endogenous information cost (i.e., subjects needed to bid for information) in

experimental decentralized asset markets. For this, the current paper needs to carefully discuss why using a fixed cost of information (ECU6 per ball) and allowing subjects to choose how many balls to draw is a better setup and can serve for your research purpose. The paper needs to discuss how market structure differs from that of Asparouhova et al. (2024) and guide the readers to understand what you expect to observe – which is same or different from the existing work. Besides, Asparouhova et al. (2024) finds the informed traders outperform the uninformed, but equally well after considering the cost they paid for acquiring information. It suggests the cost of information is rational. Can the current paper somehow show whether a certain information acquisition choice is rational for traders? [in your theoretical discussion section]; and whether subjects behaved in the expected way in experimental sessions? [in your results section]

2. **Theoretical discussions:** the current manuscript lacks theoretical discussions and fails to explain experimental design in detail. A lot of discussions and arguments are around Zhu (2014)'s idea of non-execution risk in dark venue. It is ok, however, I believe the most important thing the paper needs to focus is NOT on the relationship between dark pool trading and market liquidity, but information acquisition. Zhu (2014) model has no information acquisition and does not prescribe whether the information acquisition would increase or decrease when a dark pool is added into trading landscape. I would suggest to:
 - 1) link the paper much closer to Ye (2024) model so to justify the experimental treatment of high/low precision signals and link the paper to the experimental setting of Asparouhova et al. (2024) which also has high/low information settings; and discuss experimental findings against the existing evidence; 2) add theoretical discussion on experimental design and explain the relevant design detail. In particular,
 - 2.1 To my understanding, subjects were not informed whether they were in the high-precision or low-precision treatment. This should be stated explicitly. In several places, the wording suggests that subjects knew their signal would be strong and therefore chose to acquire information, which could be misleading. Please revise to clarify that subjects did not know the precision level when making their decisions.
 - 2.2 To my understanding, the current paper uses these two settings – high/low information precision- to generate data for “moderately informed” vs “strongly informed” individuals (defined by their posterior beliefs) so that their trading behaviour would be tracked and the corresponding market outcome would be observed.

- 2.2.1 It would be helpful to present some statistic on how many observations for each group of individuals. That is, add the columns of “No. Observations” in Table IV.
 - 2.2.2 If high/low previous information setting is your treatment, discuss what you expect to observe about traders’ behaviours or market outcome across high/low-precision sessions. This is necessary for experimental studies.
 - 2.2.3 Discuss what you expect to observe about individual trading behaviours conditional on their posterior strengths. This is crucial in your design.
3. **“Sorting effect”:** “Sorting” is not well defined nor explained. The wording only shows up in Introduction and Discussions, not being mentioned anywhere else.
 - 3.1 Ye (2024)’s sorting effect refers to the phenomenon that traders of various signal precisions would have different venue choices. But, in your experiment, each period, the information precision is pre-determined, and it is a treatment variable. I am not quite sure whether your sorting effect is in line with Ye (2024). Or do you mean Table V is the evidence of sorting effect? Not discussed, though.
 - 3.2 The term “sorting” is not widely used in the existing literature. It can potentially create some confusions, too. How about removing it in the title? Say, “Information Acquisition and Price Informativeness in Experimental Dark Markets”?
 - 3.3 I have difficulties with the discussions around *crowding-out effect*, *information acquisition effect* and *sorting effect* on page 4 and 5. “*As the dark market participation goes up, the crowding out effect grows stronger and eventually exceed the positive effects.*” What does it mean? The “crowding-out effect” is something sourced from market fragmentation: when market is fragmented, the volume in one venue is less than when the market is concentrated. “Information acquisition” is traders’ action, not an effect. Then the sorting effect seems to lead traders of strong signals to use the lit more than the dark pool. Well, do you try to argue that the reduction in trading volume on the lit exchange caused by market fragmentation could be mitigated if traders with strong signals choose to trade in the lit market rather than in the dark pool? Do you think these discussions would help readers to understand why the information acquisition is higher in the presence of dark pool?
 4. **Experiment design:**
 - 4.1 **The fixed amount of information cost?** In your design, it cost subjects ECU6 to know the color of a ball, and they need to decide on how many balls to draw. I guess there exists an optimal choice of the information acquisition. Such a number should

be the same across the high/low precision sessions because subjects were not informed of which session they sat in. You could argue theoretically this is better than the information auction because it is a discrete choice – which leaves room for profit and thus motivates to trade on information. In-depth discussions on how the experimental settings fit in the hypothesis testing and what to expect in experiments would enhance the quality of the paper.

4.2 Information acquisition: if the balls were drawn independently, why did you impose a cap of 10 balls for information acquisition? Maybe it is a number far beyond the optimal choice, but helps you to reduce the chance of subjects making irrational choice and ruining the experiments? In results section, I wish to see some statistics of information acquisition decisions, such as how many balls the subjects actually chose to pay to see.

4.3 Role of traders who are not offered to acquire information:

4.3.1 I guess the design is to ensure there would surely have some uninformed traders in markets. I would encourage the author(s) to explain why in the paper.

4.3.2 These traders do not have liquidity obligation. What motivated them to trade? Extreme risk aversion could lead to a no-trade situation, right? If so, only informed traders trade against each other. Please explain how informed and uninformed interacted with each other theoretically and empirically. For example, given the trading activities seem to be mainly driven by various information signals, it would be helpful to include some statistics at individual trader level conditional on the signal they receive. The current analysis is conducted at the “group” level – weakly informed, strongly informed group.

5. Re-arrange the results: the current manuscript contains about 20 pages of results (page 14 – 34). Please go through them and make some adjustments based on their importance and your flow of arguments. I would think Result 6 in Section D is what your paper focuses on.

5.1 Result 1: I personally would lower the priority of this. Other papers (e.g., Hendershott et al., 2022) have documented similar patterns.

5.2 Result 2: one of the advantages of experimental method is to allow us to observe the true value (fully revealing price). This is one of the “must-show” results in experimental asset market studies. But you may need to think through where to present the results.

5.3 Section C on information efficiency: would it be possible to show price informativeness over time? It will demonstrate how quickly the information has impounded into price. My intuition is: traders of strong signals, assuming they are not the only informed in the market, would be eager to capitalize their informational advantage. By doing so (please check if their orders are aggressive), the information revelation would happen earlier. Please consider a graph with x-axis being time and y-axis the measure of price efficiency or forecast error or similar. Say, Figure 1 & 2 in Bossearts et al. (2013).

5.4 Section D: add more analysis on information acquisition. Say, statistics on how many individuals paid to acquire signals and how they performed. Naturally if they benefit from information acquisition, they would continue doing so through an experimental session. If so, how many of such individuals can you identify in each of sessions? If they did not make money, did they stop information acquisition? Their choice of information acquisition, to my understanding, would benchmark against the uninformed trader's payoff.

Other comments:

6. "traders with stronger signals find it extremely difficult to execute their order in the dark pool" Can you please show some statistics, such as fill rate, time until execution to demonstrate your point, if you decide to show these in the main text? See also Table 6 of Hendershott et al. (2022).
7. What's the incentive for subjects who cannot acquire information? They have endowments 600ECU + 40 Units of assets. If they do not do anything, they would get either 600ECU or 1000ECU which is SGD12 or SGD20. Adding the showup fee, a subject who does not do anything would have SGD14 or 22 (with the average being SGD18). For other traders, given trading is a zero-sum game, the average payoff should be SGD18, too. This does not match with Footnote 6 which reports payoffs ranged from \$9 - \$30 (the average being 19.5).
8. Market efficiency is measured against the "full-revealing price" (please clarify whether it is the same as "true value" since both show up in your writing). Would you be able to demonstrate how a trader's order price or transaction price relates to his posterior price conditional on information? This helps us understand if subjects behaved in the predicted direction.

9. Define all empirical variables clearly before Results section. Alternatively, provide a table which contains all empirical measures and their definitions.
10. Table V, are two dependent variables significantly different across treatment? If so, please add statistics as a panel in the table.
11. Footnote 26, Hendershott et al. (2023) is probably a better comparison than Bloomfield et al. (2015) since it has the lit-dark markets as yours.
12. Reference list needs to be tidied up. would encourage the authors to cut down the number of papers. Most of the papers are cited only once. While some key papers are missed out.
13. Editing issues (not exhaustive list):
 - 13.1 Page 12, “between-subjects design”
 - 13.2 Page 13, what is MUDA? If being used only once or twice, change into the full name as opposed to the abbreviation.
 - 13.3 Several places mentioning “theory” fail to clearly point to what theory. For example, “in line with theory”, “suggested by theory” “the predictions of the theory” What theories do author(s) refer to?
 - 13.4 Abstract “In line with theory (what?), investors with stronger signals use the lit exchange relatively more (than what?)” My understanding is: investors use the lit venue more than the dark pool; and this is true when traders have high precision signals.
 - 13.5 Page 2, 2nd paragraph, “Several of these dark pools are...” In fact, dark pools, classified by the operator, are exchange-operated dark pools (such as Center Point in Australia, Turquoise in UK) and broker-operated dark pools; by execution algorithms, dark pools can be considered as one-sided market (which uses mid-point execution) or two-sided market (which uses price-time priority rule). The current experimental setting, to my understanding, uses the mid-point execution, which is in line with most of models such as Zhu (2014), Werner et al. (2023) and Ye (2024). If the author explains the details here, no need for footnote 17.
 - 13.6 Figure 2 & 3, there are two “dashed lines” and using a different line pattern would be better.
 - 13.7 Page 24, why “fully revealing price” is 10? It would be better if it is being discussed in a newly added theoretical discussion section.
 - 13.8 Please check through your footnotes. There are too many: either delete (if minor) or incorporate some into the main text (if important), and only keep those sit in the

middle as footnote. For example, delete footnote 20, incorporate footnote 21, and keep footnote 22. Incorporate footnote 7.

13.9 Page 39, the first sentence, why? And the second sentence, why would heterogeneous endowments be an interesting setting? Essentially the last paragraph does not really provide any in-depth thoughts. Please revise it.

References

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